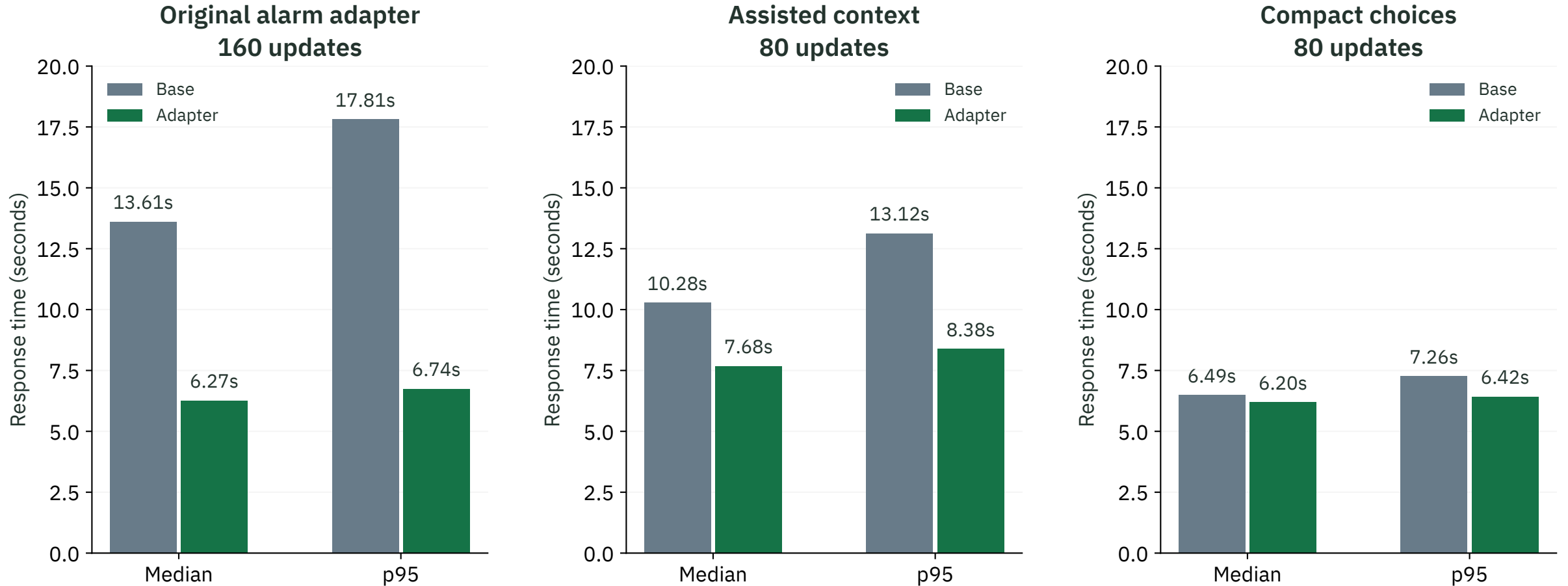


Training shortened response time

Matched local Qwen3-4B / MLX within each panel. Same 32 cases, sequential inference; first response excluded (31 warm).

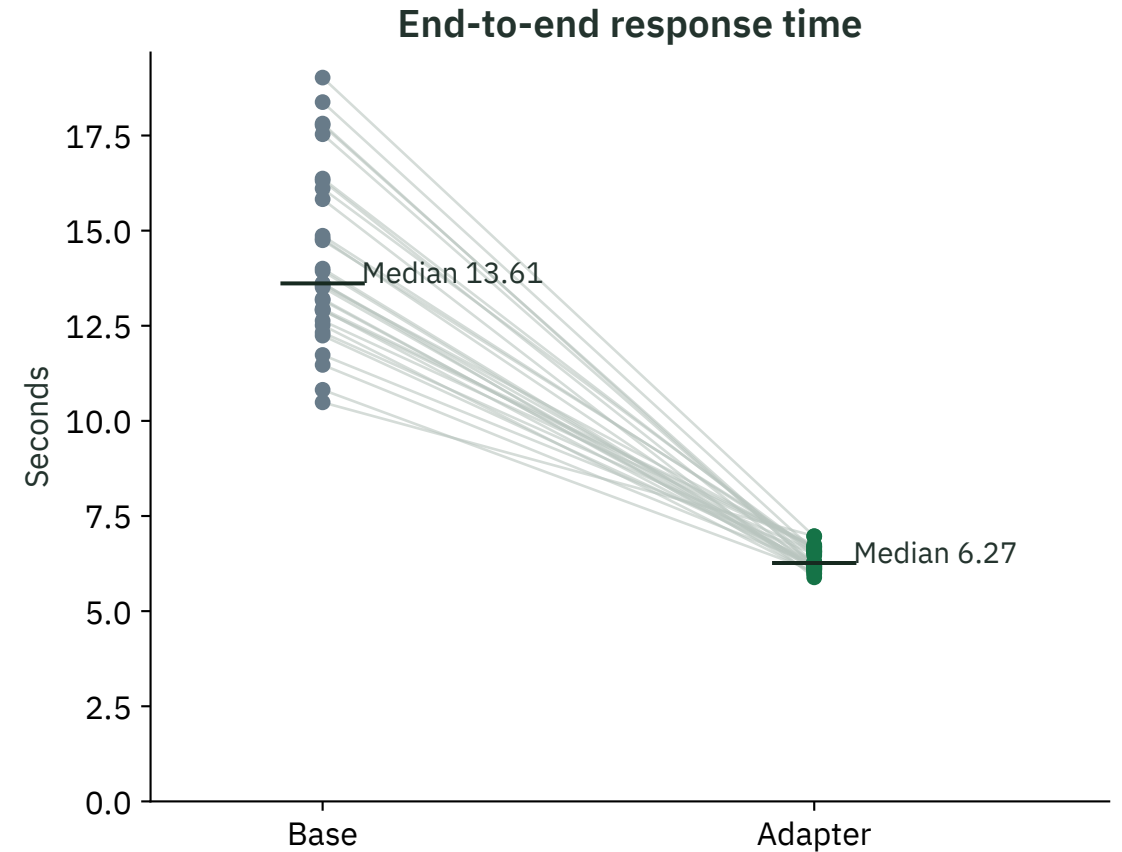
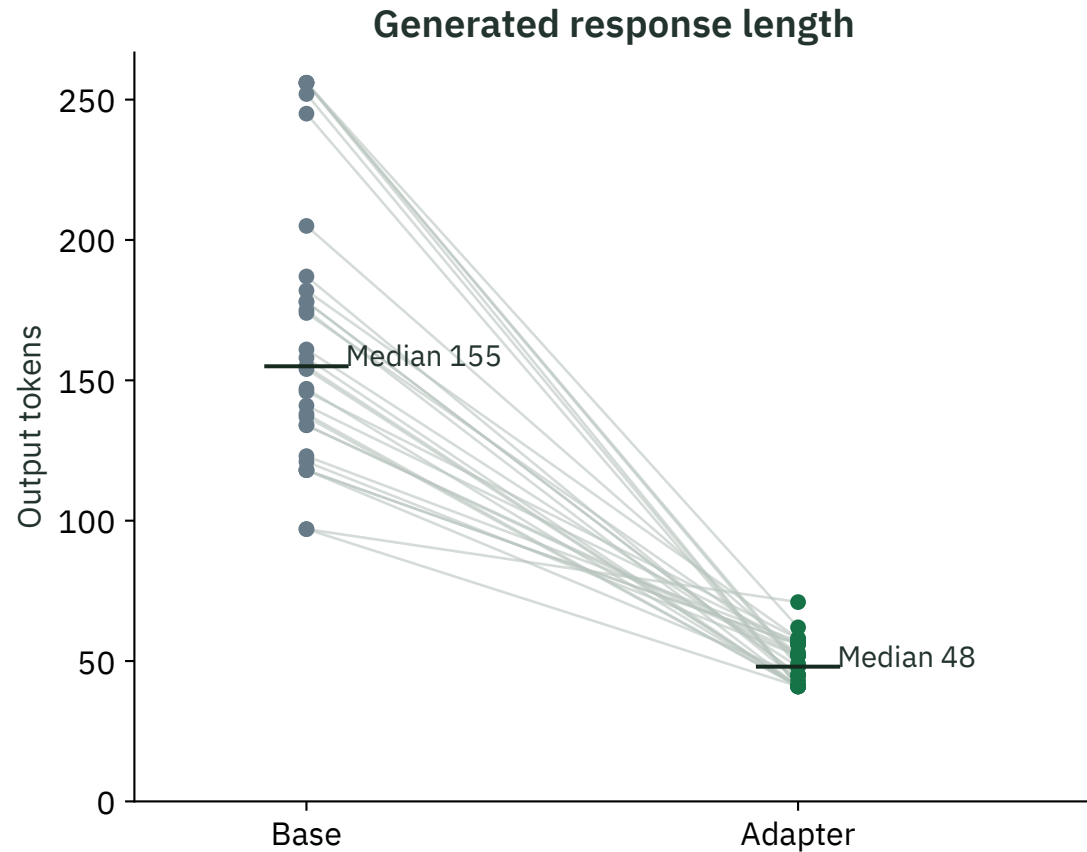


Original adapter: 54.0% lower median and 62.2% lower p95 response time.

Panels are different experiments, not a chronological speed benchmark. Pilot adapters escalated every case and remain unapproved.

Shorter responses, faster completion

Original 160-update adapter: paired local measurements, identical inputs and decoding limits; 31 warm cases.

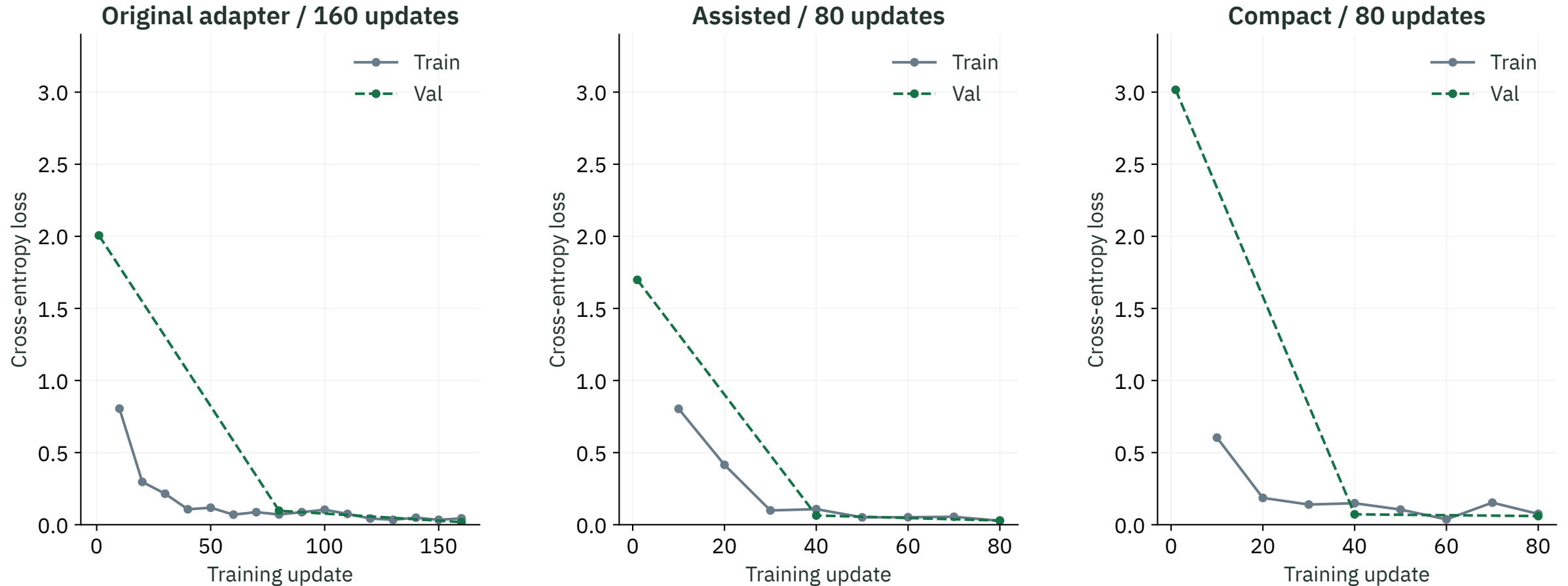


Each connector joins the same case. Black marks show medians.

Elapsed response time includes prompt processing and generation; these records do not isolate decoding tokens/second.

Loss improved; control quality still needs testing

Training logs. Validation loss used four examples at each recorded checkpoint; this is not the full decision benchmark.

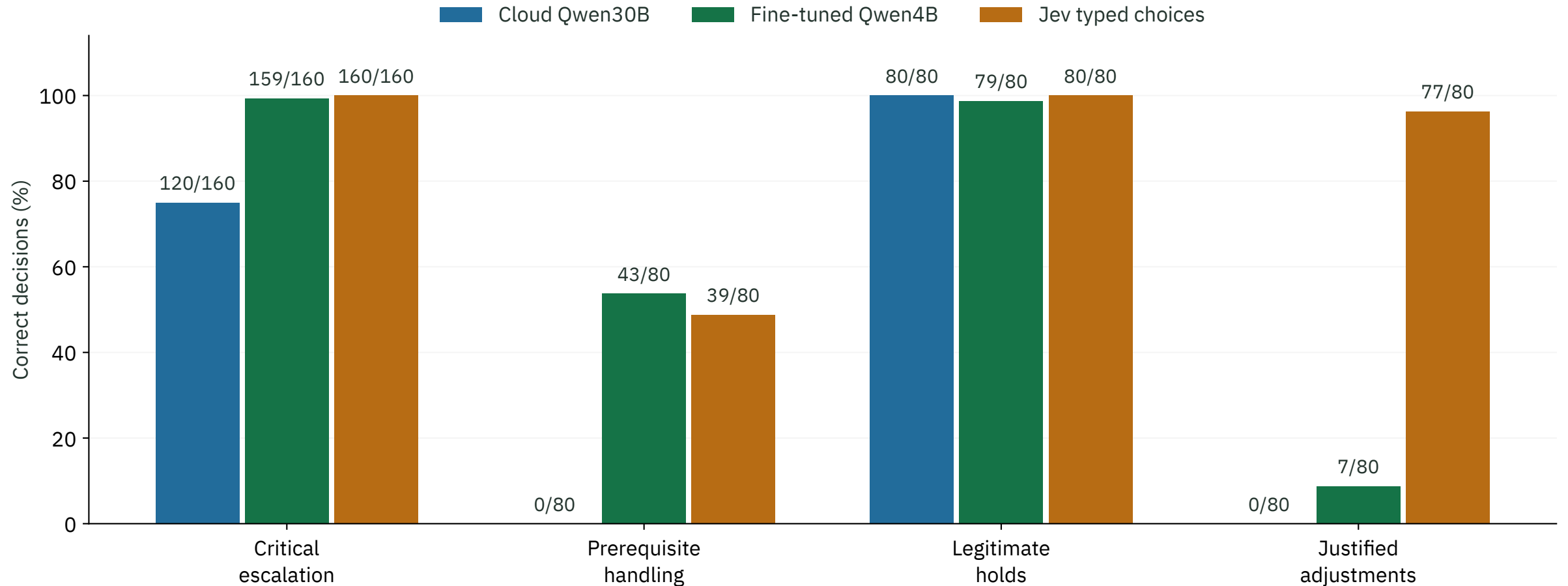


Lower loss measures fit to the training objective. It does not establish correct hold/adjust behaviour.

Both later adapters matched the 16/32 always-escalate baseline despite large loss reductions.

Shared cases: where each system succeeded

400 development cases. The adapter was selected using this set; Jev receives code-defined choices and evidence assistance.

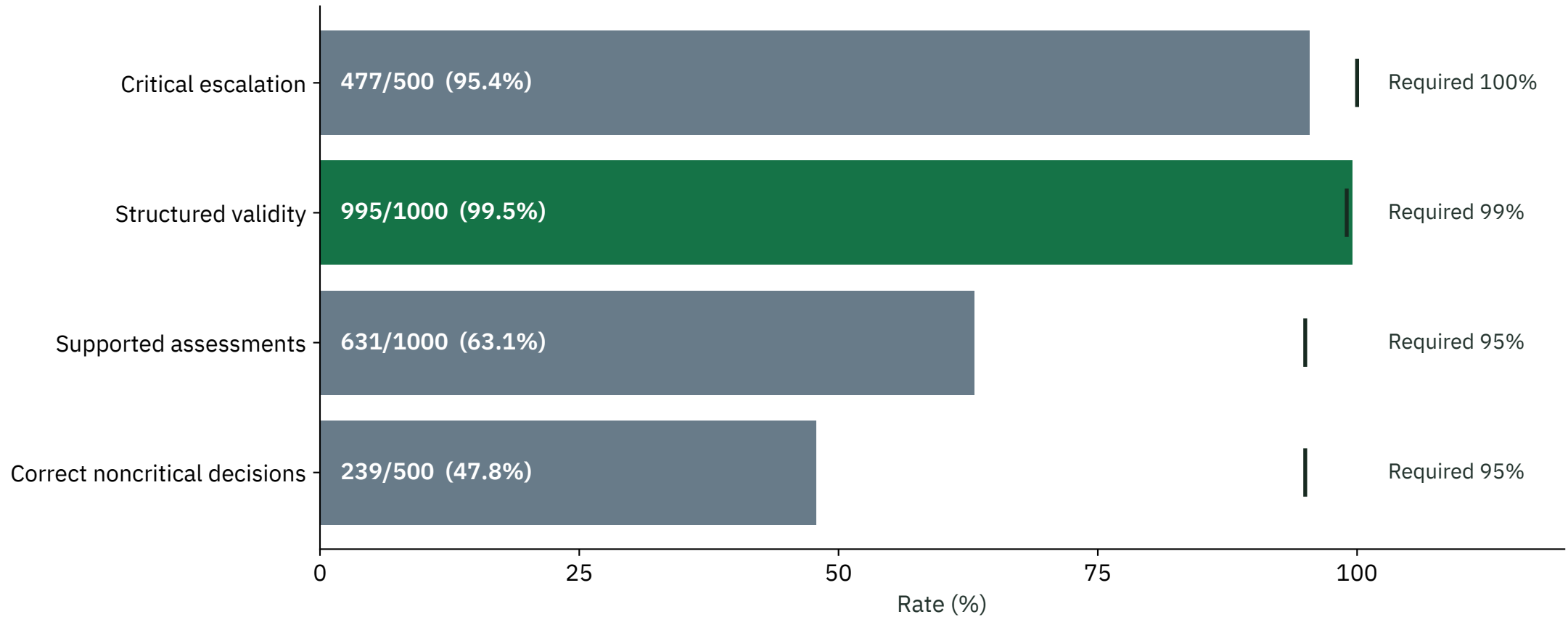


Overall correct: Cloud Qwen30B 200/400; fine-tuned Qwen4B 288/400; Jev 356/400.

This is a system comparison across different interfaces, not a controlled comparison of model intelligence.

Locked test: the adapter did not qualify

1,000 held-out cases, including 500 critical cases. Original fine-tuned Qwen3-4B; benchmark proposals did not actuate the plant.



Black marks indicate the predeclared threshold. There were also two fabricated-reference errors (required: zero).

Failure of an offline escalation criterion does not mean an unsafe command was executed.

Later pilots: escalation improved, useful control did not

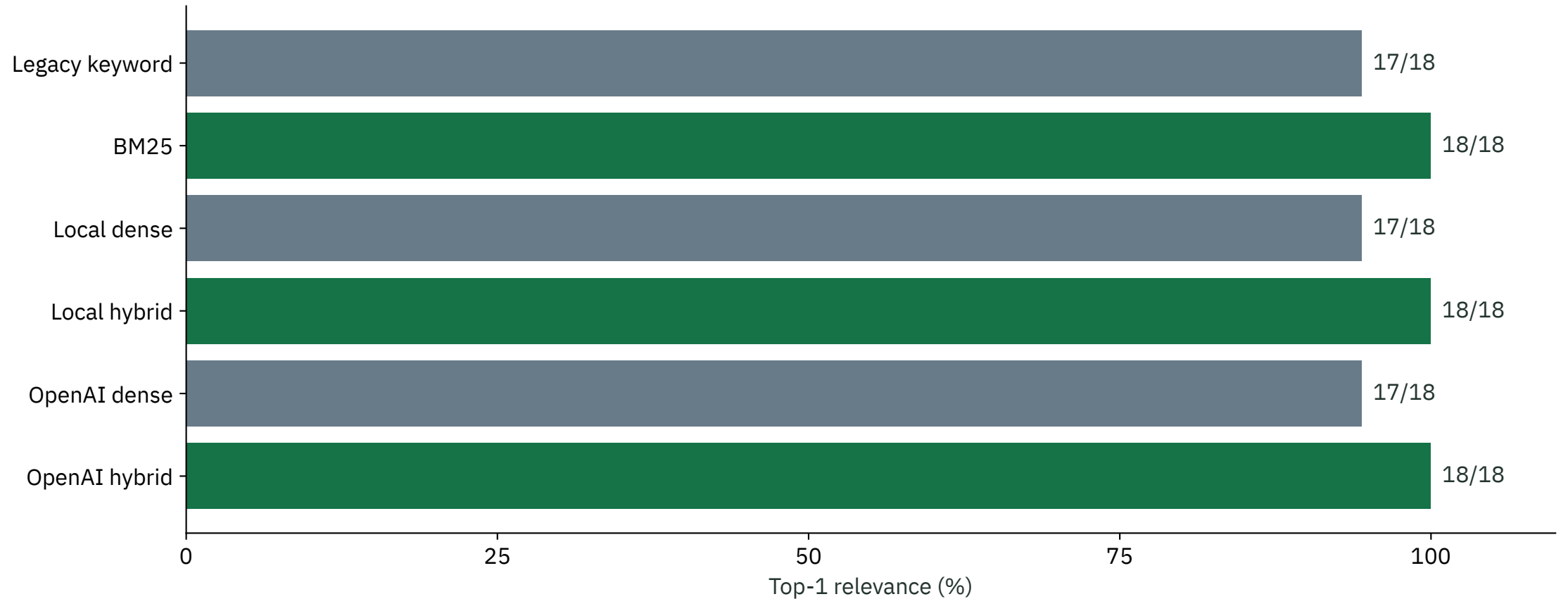
Same 32 fresh development cases (eight per category). New adapters: 80 QLoRA updates each; no new locked test.

Assisted base	2/8	5/8	0/8	0/8
Previous adapter + assistance	6/8	1/8	8/8	0/8
New assisted adapter	8/8	8/8	0/8	0/8
Compact base	4/8	0/8	8/8	0/8
New compact adapter	8/8	8/8	0/8	0/8
	Critical escalation	Prerequisite handling	Legitimate holds	Justified adjustments

Both new adapters escalated all 32 cases: correct on critical/prerequisite cases, incorrect on holds/adjustments. Their overall 16/32 equals the always-escalate baseline. Faster output alone is not a successful control policy.

Plant retrieval: a small, recorded relevance check

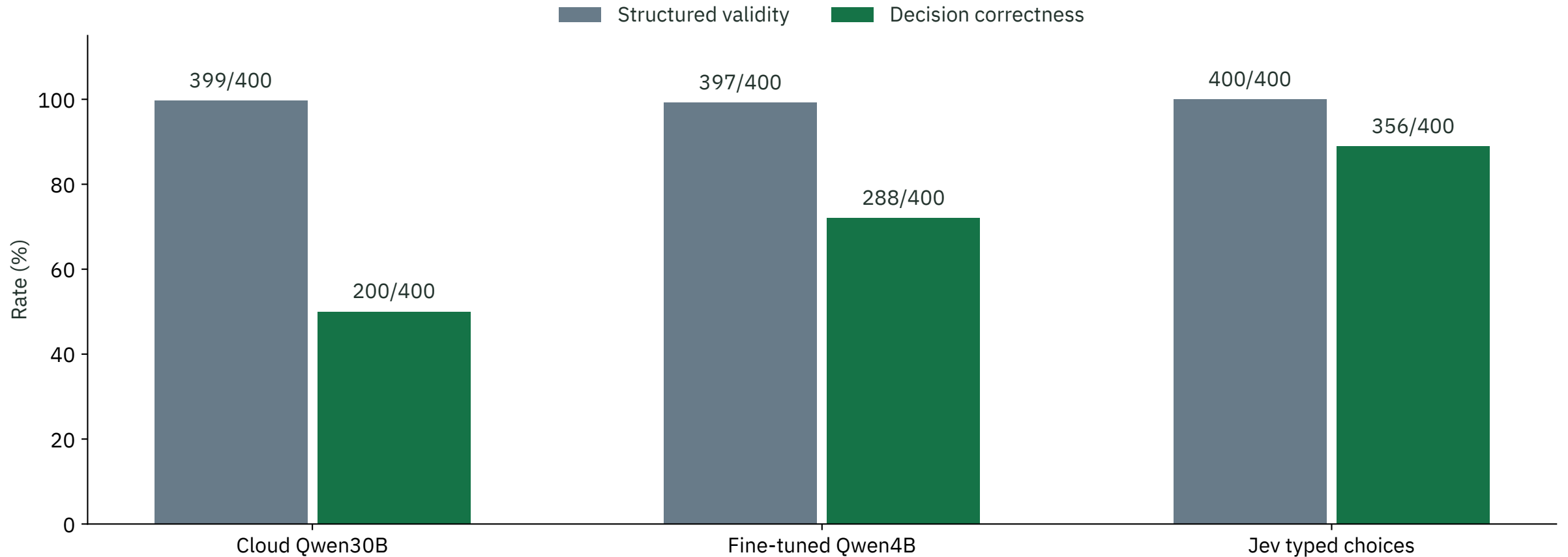
18 developer-authored queries spanning water, grid and nuclear. Metric: expected SOP record ranked first.



BM25 and both hybrid methods found the expected first result in 18/18 cases; dense-only and legacy keyword found 17/18. These relevance results do not measure model decision accuracy or plant recovery.

Valid JSON is not the same as a correct decision

Shared 400 development cases; all successful-response records retained. Same source records as chart 04.



The cloud model produced valid structured responses in 399/400 cases, but correct decisions in 200/400. Format validation and independent control checks address different problems.